

Charging/Discharging Dynamics in Two-Dimensional Titanium Carbide (MXene) Slit Nanopore: Insights from molecular dynamic study



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ABSTRACT

While most theoretical studies on slit nanopore electrodes are limited in their static properties, understanding the dynamic charge storage is crucial for promoting their electrochemical performance in the field of energy storage systems. In this article, the charging/discharging dynamics in MXene ($\text{Ti}_3\text{C}_2(\text{OH})_2$) electrode featuring intrinsic slit nanopore with room temperature ionic liquid are investigated by molecular dynamic simulation. The galvanostatic cycling is modeled to help to understand the dynamic charge storage mechanism. The simulation results show that the electrolyte ions spontaneously wet the $\text{Ti}_3\text{C}_2(\text{OH})_2$ slit nanopore in the absence of external potential, even at 0.7 nm layer distance pore case. An intriguing electroneutrality breakdown and faster diffusion property is observed inside the nanopore. During charge/discharge process, the charge storage takes place by a fast counter-ion insertion and counter-ion/co-ion exchange. Besides, the spatial distribution of ions and the orientation of the cation dipoles inside the pore vary cyclically along with the charge/discharge process, which helps to screen the external electric potential. Having elucidated the charge mechanism, we investigate the layer distance factor and show that the ions exhibit more significantly acceleration and more ordered dipole orientation inside the narrower pore. This will help $\text{Ti}_3\text{C}_2(\text{OH})_2$ electrode with narrower pore exhibit better power performance.

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1. Introduction

Recently, MXene, representing a new family of two dimensional transition metal carbides and carbonitrides, have been extensively exploited due to its unique physicochemical properties, since its discover in 2011 [1]. The metallic conductivity, intrinsic hydrophilic nature, coupled with good mechanical properties, have made MXene an excellent candidate as energy storage materials [2–5]. A variety of ions, containing alkali ions, alkali earth ions and organic ions, are observed intercalating between the atomic layers of MXene nanosheets rapidly, accompanied by a deformation of MXene nanosheets [6]. The rapid ionic intercalation happens with nearly no diffusion limitations and leads to a high volumetric/gravimetric capacitance simultaneously [7–9], posing a “paradox” contrast to traditional charging mechanism of supercapacitors and batteries.

A molecular scale understanding of the electrode/electrolyte interface in such nanoconfining environment is indispensable to reveal the mechanisms of charge storage and develop more

efficient intercalation strategies for the MXene and other two-dimensional layered materials. The electrolyte ions in the slit nanopore suffer qualitatively different confining environments, comparing with bulk electrolyte [10,11]. And the electrical double layers formed at the solid-liquid interface prefer transition from multilayer to monolayer [12], making the charge storage more efficiently [13]. Furthermore, the dynamic responses of electrolyte ions, including the real time response of ionic transfer, exchange and electrical double layer reconstruction during electrochemical charge/discharge process are also key aspects.

Experimentally, *in situ* NMR [14,15], infrared spectroscopy [16,17] and X-ray diffraction [18] techniques have been recently performed to characterize the dynamic response to temporal variations in applied voltage. By *in situ* X-ray diffraction study, a dynamic expansion and shrinkage of the slit nanopore during electrochemical process is observed in MXene nanosheets [19], and also in MoS_2 layers [20]. Griffin et al.'s *in situ* NMR study confirms that the ions in the in-pore environment are primarily responsible for the charge storage [15], and ions are in dynamic equilibrium between the in- and ex- pore environments with an exchange rate in the order of tens of Hz [21]. Prehal et al.'s study shows that the global number of ions within the carbon nanopore does not vary during charge and discharge process, and the electric

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double layer formation is accompanied by a combination of counter-ion directly at the wall [22].

The theoretical atomistic simulation study, could gain several important insights into the dynamic response of charge storage, and complements the restrictions of the *in situ* measurements in understanding the underlying charge storage mechanism, which is complicated by the complex interplay of several molecular scale phenomenon [23]. However, the charging dynamic simulations of nanopores to date are concentrated on carbon-based materials [24–26]. The full atomistic simulation of MXene nanopore upon charge/discharge is rare and urgently needed. The objective of this work is to give an atomic insight into the rapid charge storage mechanism in the slit MXene nanopore under galvanostatic charge/discharge conditions. Here, the most studied MXene Ti_3C_2 nanolaminate [27], decorating with the primarily surface terminal hydroxy ($\text{Ti}_3\text{C}_2(\text{OH})_2$), is investigated in this study. First, we describe the details of short range interaction and charge polarization of the electrode models under investigation. Then we investigate the charging/discharging mechanism with the help of in-pore ionic transfer and exchange process, the in-plane density and charge distribution of the in-pore ions and ionic orientation analysis.

2. Model and Method

In the simulation system shown in Fig. 1, the anode and cathode electrodes are separately represented with double layer $\text{Ti}_3\text{C}_2(\text{OH})_2$ nanosheets, each containing 5382 atoms. The room temperature ionic liquid (RTIL), an emerging and promising electrolyte for the electrochemistry supercapacitors, is employed. An RTIL usually consists of an organic or an inorganic anion and an asymmetric organic cation, and the RTIL's chemical and physical properties can be highly tunable due to the vast variety of combinations of anions and cations. The imidazolium-based RTILs can provide high ionic conductivity and wide electrochemical stable potential window, simultaneously, which is benefit for increasing the power density and energy density. In addition, anions of RTILs play an important role in their hydrophobic-hydrophilic properties. Considering the above experimental factors and simulation efficiency and feasibility, the imidazolium-based cation $[\text{EMIM}]^+$ and hydrophobic anion $[\text{PF}_6]^-$ are selected as the electrolyte [28,29]. The 1260 $[\text{EMIM}]^+[\text{PF}_6]^-$ electrolyte ion pairs are randomly placed surround the $\text{Ti}_3\text{C}_2(\text{OH})_2$ nanosheets initially. The validity of such simulation system has been proved in previous studies [26,30]. Eight slabs are placed at the entrance of the nanopore as a physical boundary to restrict the electrolyte ions approaching the outside surface of $\text{Ti}_3\text{C}_2(\text{OH})_2$. The size of the orthorhombic simulation cell is $24.0 \times 4.0 \times 6.0 \text{ nm}^3$ and the periodic boundary condition is set in all three directions. The length of the slit nanopore and the distance between the two electrodes are both set to 6.0 nm, which is large enough to ensure the interaction between the two electrodes is negligible. Three interlayer distances (viz. $d = 0.7 \text{ nm}$, 1.0 nm , 1.4 nm) are set to investigate the charge storage mechanism

at different expansion situation when electrolyte ions inserted into the nanopore.

The OPLS-AA force field is utilized in this work to determine the interaction between the atoms and molecules [31], whose validity has been verified in previous studies [32,33]. The detailed force field and partial atomic charge parameters for models in our simulations are summarized in Table 1 below. The force field parameters of $[\text{EMIM}]^+[\text{PF}_6]^-$ ions are taken from the original OPLS-AA force field set [34,35]. Due to the recent discovery of MXene, an accurate expression for charge polarization and short range interaction potentials directly fitted to the interaction energy in $\text{Ti}_3\text{C}_2(\text{OH})_2$ nanosheets have not yet been established. In this work, the local geometric structures of the $\text{Ti}_3\text{C}_2(\text{OH})_2$ nanosheets are determined by the density functional theory calculations [36], and the partial atomic charge parameters of $\text{Ti}_3\text{C}_2(\text{OH})_2$ atoms are obtained based on Mulliken population analysis [37]. Considering the analogous chemical surface environment between $\text{Ti}_3\text{C}_2(\text{OH})_2$ and $\text{TiO}_2\text{-OH}$, it may be reasonable porting their Lennard-Jones (LJ) parameters to represent the short range vDW interaction, and the LJ parameters of Ti, O and H atoms are following the previous work [38]. The LJ parameters of C atom are the same as that for RTIL model. Considering the main deformation of MXene nanosheets is the expansion and shrinkage of MXene layers, which is due to the interaction and de-interaction of the electrolyte ions [19,39]. And the amplitude of expansion and shrinkage is slight, approximately 5%–10% of the interlayer distance. The five layer Ti_3C_2 slabs are kept frozen, and the terminal hydroxyl groups ($-\text{OH}$) are flexible during the simulations. The Ti-O and O-H bond parameters and Ti-O-H angle parameters are extracted by fitting to the potential surface obtained via density functional theory calculations. The fitting details are provided in the supporting information. For different

Table 1
Force field parameters.

LJ and charge coefficients			
Element	$\varepsilon/\text{kcal.mol}^{-1}$	σ/nm	partial charge/e ⁻
Ti-inner	0.6087	0.19565	0.64
Ti-outer	0.6087	0.19565	0.88
C	0.0660	0.35000	−0.76
O	0.1554	0.31656	−0.79
H	0.0000	0.00000	0.35
Bond coefficients			
	$k/\text{kcal.nm}^{-1}$		r_0/nm
Ti-O	1244.9		0.2189
O-H	5541.35		0.0973
Angle coefficients			
	$k/\text{kcal.}\theta^{-1}$		$\theta/\text{degrees}$
Ti-O-H	7.062		125.87

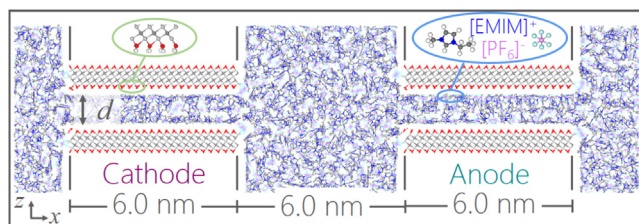


Fig. 1. Schematic representation of the simulation system. Left and right double layer $\text{Ti}_3\text{C}_2(\text{OH})_2$ nanosheets denote the cathode and anode, respectively. Blue and magenta lines denote the $[\text{EMIM}]^+$ and $[\text{PF}_6]^-$ ions, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

kinds of atoms, the Lorentz-Berthelot combination rule, i.e. $\epsilon_{ij} = \sqrt{\epsilon_{ii}\epsilon_{jj}}$ and $\sigma_{ij} = (\sigma_{ii} + \sigma_{jj})/2$, is employed to obtain their Lennard-Jones interactions [40].

All simulations are performed in the NVT ensemble with the target temperature 450 K, using the Lammmps molecular dynamics (MD) package [41]. A time step of 1 fs is utilized. Spherical cutoff radii for the Van der Waals and Coulomb interactions are set to 1.0 nm and 1.6 nm, respectively. The long-range electrostatic interactions are calculated via the particle-particle particle-mesh scheme (pppm) in k-space. In order to simulate the ion dynamic response and restructure of the electric double layer (EDL) formed inside the nanopore during the galvanostatic cycling process, the simulation systems are firstly equilibrated with zero net charge until the system reaches equilibrium state. Then the charge process starts, a small partial charge is continuously add (remove) to the anode (cathode) until the net charge density on the electrode reaches 90 $\mu\text{C}/\text{cm}^2$, and the charge current density is set as 200 kA/cm^2 . Then the discharge process follows with removing (adding) the net charge on anode (cathode) at the same rate. We note that the current density is orders of magnitude higher than that common used in experimental studies systems. In our simulations, the targeted temperature 450 K helps increase the diffusion coefficient of the ions (see below, ~ 1 –2 order of magnitude larger than that of typical values experimental observed). Due to the large ionic diffusion coefficient, the formation and relaxation of EDLs is fast. In addition, the length of the simulated nanopore is orders of magnitude shorter than the common experimental studied systems. Hence, the nanopores retain significant charge storage capability under high current density, and this high current density still allow us to investigate the mechanism of charging/discharging dynamic in nanopores efficiently [24,26]. In order to acquire reliable statistics, four charge/discharge cycles are achieved in one simulation, and each simulation case is repeated 10 times with independent initial configurations.

3. Results and Discussion

3.1. Number evolution of electrolytes ions and net-charge during charge/discharge process

First, we focus on the ion insertion into the nanopore at neutral state. The number of $[\text{EMIM}]^+$ and $[\text{PF}_6]^-$ insertion into the neutral pore, marked as n_E^n and n_P^n respectively, the total-ion n_T^n ($n_T^n = n_E^n + n_P^n$) and net-charge n_q^n ($n_q^n = n_E^n - n_P^n$) are shown in Fig. 2. We could find that the electrolyte ions could insert the nanopore spontaneously, in all three distance cases. The polar surface of the $\text{Ti}_3\text{C}_2(\text{OH})_2$ sheet make the nanopore ionophilic, which is benefit for effective utilization of the inner pore surface. The total-ion numbers increase as the volume of the pore increase, and the density of electrolytes remains almost constant and nearly equal to the bulk density. Noting the net-charge number, an intriguing phenomenon beyond our common sense is observed. An

electroneutrality breakdown is achieved, due to the slight higher quantity of $[\text{EMIM}]^+$ than $[\text{PF}_6]^-$ in the nanopore. The positive net-charge of the electrolytes are ~ 4.4 e, ~ 2.8 e and ~ 1.8 e in $d = 0.7$, 1.0 and 1.4 nm pores, a slight decrease as the confinement decrease. In this nanoconfinement condition, due to the identical charge of the cation and anion, the columbic interaction of cation-electrode and anion-electrode should be equal. The asymmetric specific adsorption interaction between anions and cations with the electrode should be responsible for such electroneutrality breakdown. Analogously, experimental NMR studies of the aqueous electrolytes in nanoconfined carbon pores observed a negative net-charge [42]. The stronger $[\text{EMIM}]^+$ -electrode specific adsorption attracts more cations into the nanopore than anions, and the net-charge in the nanopore becomes positive. On the contrary, the specific adsorption interaction of anion-electrode is sometimes stronger than cation-electrode in a carbon-based aqueous system [43], and so a net negative charge is observed.

During the galvanostatic cycling simulation, the net charge variations on the cathode and anode are denoted with the blue wavy line in Fig. 3(a) and (b), respectively. The excess net charges induce a potential difference between the cathode and anode, and lead a directional migration of the electrolyte ions. The counting net-charge of electrolyte in the nanopore varies following the electrode net-charge variation at nearly same rate, as shown in Fig. 3(a) and (b). The net-charge of electrolyte in the nanopore reach maximum values of 71.9 e, 80.6 e, 91.1 e in $d = 0.7$, 1.0, 1.4 cathode cases, and minimum values of -79.5 e, -84.6 e, -87.9 e in $d = 0.7$, 1.0, 1.4 anode cases. There is no remarkable increase of the peak value when the layer distance increase from 0.7 nm to 1.4 nm. Comparing with the 135 e net-charge on the electrode, we can see that the excess electrode charge is not totally screened by the opposite net-charge in the electrolyte. Some other screen mechanism, except the rapid ion diffusion, may exist to help to screen the external electrostatic field, and will be discussed below.

The accumulation mechanism of the net-charge of the electrolyte in-pore can be divided into two parts: the counter-ion adsorption and the counter-ion/co-ion exchange. Under the electrostatic potential, the counter-ions are adsorbed into the pore and the co-ions are repelled from the pore, simultaneously. The number evolution of $[\text{EMIM}]^+$ and $[\text{PF}_6]^-$ in $d = 1.0$ nm cathode and anode pore during charge/discharge process are represented in Fig. 3(c) and (d), as a representative. It can be seen that both in cathode and anode pore, the counter-ion adsorption goes faster than the co-ion exclusion during the cycling process. When charge stops and discharge begins, the counter-ion adsorption stops almost instantly, while the co-ion represents a delay response. When discharge finishes, the electrode sheet return to neutral, the $[\text{PF}_6]^-$ number remains deviated from its equilibrium values, while the $[\text{EMIM}]^+$ number returns to the equilibrium value. This different response of $[\text{EMIM}]^+$ and $[\text{PF}_6]^-$ is due to the different specific interaction with the electrode sheet and different diffusion properties.

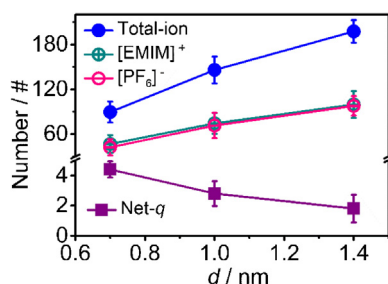


Fig. 2. Numbers of $[\text{EMIM}]^+$, $[\text{PF}_6]^-$, Total-ion and Net-charge in the neutral nanopore as a function of nanopore distance (d).

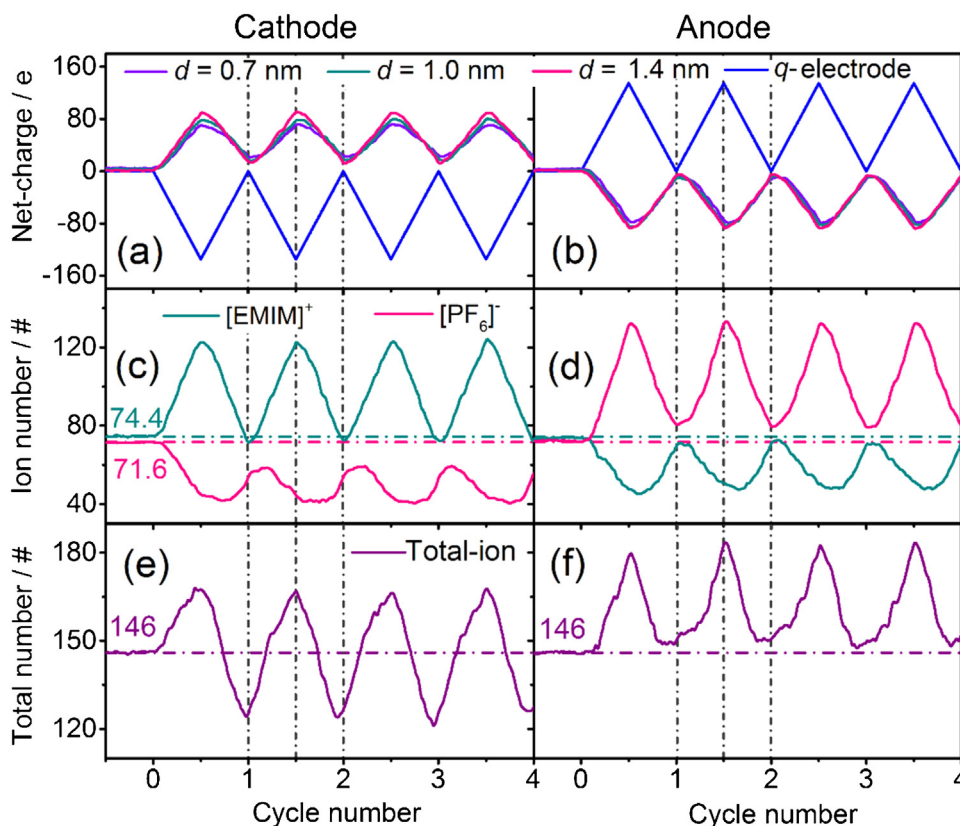


Fig. 3. Number evolution of net-charge in electrode and electrolyte in cathode (a) and anode (b) pore with different distance ($d = 0.7, 1.0, 1.4$ nm). Number evolution of $[EMIM]^+$, $[PF_6]^-$ in $d = 1.0$ nm cathode (c) and anode (d) pore. Number evolution of total ion in $d = 1.0$ nm cathode (e) and anode (f) pore. The vertical dashed lines denote the second charge/discharge cycle. The horizontal dashed lines denote the equilibrium value of the ionic numbers in the neutral pore. The profiles located before cycle number 0 denote the neutral state.

The asymmetric number evolution of cations and anions leads to a different response model of total ionic number in cathode (Fig. 3(e)) and anode (Fig. 3(f)) pore. In cathode pore, the total ionic number oscillates up and down by the equilibrium value, while an oscillation beyond the equilibrium value happens in the anode pore. Both two kind of ionic number oscillation would induce a volumetric expansion and shrinkage between the electrode layers, which is comparable to the analogous volumetric variation in previous experimental studies [19,44]. This volumetric variation during charge/discharge process may be adverse to the stability of the electrode and its cycle life, and need further investigation by both theoretical and experimental study.

3.2. Ion diffusion evolution during charge/discharge process

The conductivity and power density depends on the diffusion of the electrolyte ions inside the pores. A real time ion diffusion is analyzed through the time evolution of mean square displacement (MSD), a measure of the average distance the ions in-pore travels, which is defined as $\langle |r_i(t_1) - r_i(t_2)|^2 \rangle$. The limiting slope of the MSD at timestep t_n is related to the diffusion coefficient $D(t_n)$:

$$D(t_n) = \frac{1}{n} \lim_{n \rightarrow \infty} \frac{d \langle |r_i(t_n) - r_i(t_{n-1})|^2 \rangle}{dt} \quad (1)$$

Where n depends on the space dimensionality, and $n = 4$ in this two dimensions nanopore.

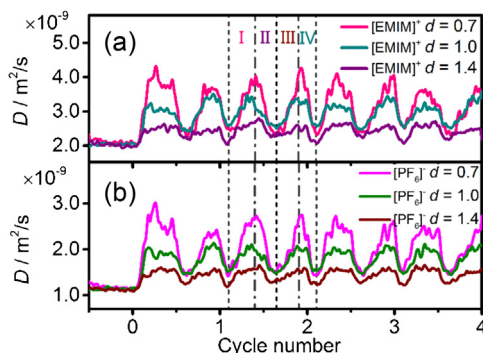


Fig. 4. Diffusion coefficient evolution of $[EMIM]^+$ and $[PF_6]^-$ in the cathode pore with different layer distance during cycling process as a representative. The vertical dashed line denote the four period in second cycle.

The calculated diffusion coefficient of $[\text{EMIM}]^+$ and $[\text{PF}_6]^-$ in the bulk region is $\sim 1.19 \times 10^{-9} \text{ m}^2/\text{s}$ and $\sim 0.67 \times 10^{-9} \text{ m}^2/\text{s}$, respectively. This diffusion coefficient is ~ 1 – 2 order of magnitude larger than that of typical values experimental observed [45,46]. This is due to the high temperature ($T=450 \text{ K}$) employed in our simulation, and the diffusion coefficient is comparable with previous theoretical studies at the same temperature [47,48]. Kislenko et al. examined the effect of temperature on interface dynamics by MD simulations, and found that increasing the temperature from 300 to 400 K had no qualitative effect on the interfacial dynamics even though the ionic diffusion coefficients increased by an order of magnitude [49]. Although the cation $[\text{EMIM}]^+$ is larger and heavier than the anion $[\text{PF}_6]^-$, the cation $[\text{EMIM}]^+$ present a faster diffusion property. This phenomenon is consistent with previous studies [45,46], and denotes that the diffusion process depends on multiple factors including volume, mass, and coordination numbers. When inserted into the nanopore, it may be expected that the confined ions present slower diffusion. However, the diffusion coefficient of both cation and anions inside the nanopore exhibit a larger value than in bulk, like other confinement systems [50,51]. Due to the confinement, electrostatically driven cation–anion attraction decreases as the interlayer distance gets smaller, and repulsion between the same cations becomes smaller, in the same way. In turn, the systematically decreased interaction among the ions under confinement accelerates the ionic diffusion [52]. The diffusion coefficient inside the three distance pores are nearly the same, and the its value are $\sim 2.11 \times 10^{-9} \text{ m}^2/\text{s}$ and $\sim 1.18 \times 10^{-9} \text{ m}^2/\text{s}$ for $[\text{EMIM}]^+$ and $[\text{PF}_6]^-$, respectively.

During charge/discharge process, it can be seen that both $[\text{EMIM}]^+$ cation and $[\text{PF}_6]^-$ anion exhibit an oscillation with frequency twice that of the charge/discharge frequency as shown in Fig. 4. This phenomenon could be referred as the competition of two opposite effect: electrostatic acceleration effect and steric restriction effect. At the initial charge stage (Period I), the interspace in the pore is relatively large and the ions interaction is smaller than electrostatic interaction. Under this condition, the ions movement increase with the charge current density. As the in-pore ions number increase (Period II) the pores become crowded,

the enhanced ions interaction and steric effect restrict the ions movement, the diffusion coefficient declines. At the discharge period (Period III and IV), the opposite process happens, inducing the diffusion coefficient increased firstly and then decreased. Furthermore, the diffusion coefficient at charged state decreases as the pore distance increases, and the diffusion coefficient of both cation and anion inside the $d=0.7 \text{ nm}$ pore appear about 2 times larger than inside the $d=1.4 \text{ nm}$ pore. It means that the $\text{Ti}_3\text{C}_2(\text{OH})_2$ nanosheets featuring 0.7 nm pore may exhibit better power density during charge/discharge process.

3.3. EDL structure evolution during charge/discharge process

Beyond information on the ion concentrations from NMR and XRT, molecular simulation allows to unravel more details about the ion arrangement information inside the pore. The density distributions of cation $[\text{EMIM}]^+$ and anion $[\text{PF}_6]^-$ in the nanopore are studied to examine the properties of EDL formed near the $\text{Ti}_3\text{C}_2(\text{OH})_2$ surface. The number density profile of $[\text{EMIM}]^+$ and $[\text{PF}_6]^-$ perpendicular to the $\text{Ti}_3\text{C}_2(\text{OH})_2$ sheet in the nanopore with different distance are presented in Fig. 5(a)(b)(c). It can be seen alternating cations and anions ordered layer structures are achieved in all three different d cases. This alternating cation and anion distribution leads to a local charge separation in the nanopore, as shown in Fig. 5(d)(e)(f), which is a key feature of the EDL. The cation $[\text{EMIM}]^+$ prefer to locate at the nearest layer to the layer sheet with a distance of $\sim 0.22 \text{ nm}$, due to a stronger short-range interaction, while the anion $[\text{PF}_6]^-$ prefer to locate at inner layer. Unlike the spatially decaying oscillation behavior exhibit near the open surface of planar electrode [33,53], we could consider the ion distribution in the slit pore as two interference “wave”, placed symmetrically around the slit center [54]. The two waves superpose to form a resultant one, with regions of constructive and destructive interference, and the resultant wave varies as the layer distance changes.

Figs. 6 and 7 represent the evolution of counter-ion and charge density profile in anode and cathode pore, respectively. We could find the EDL exhibiting cyclical rearrangement to help to screen the external force field during charge/discharge cycling process. The

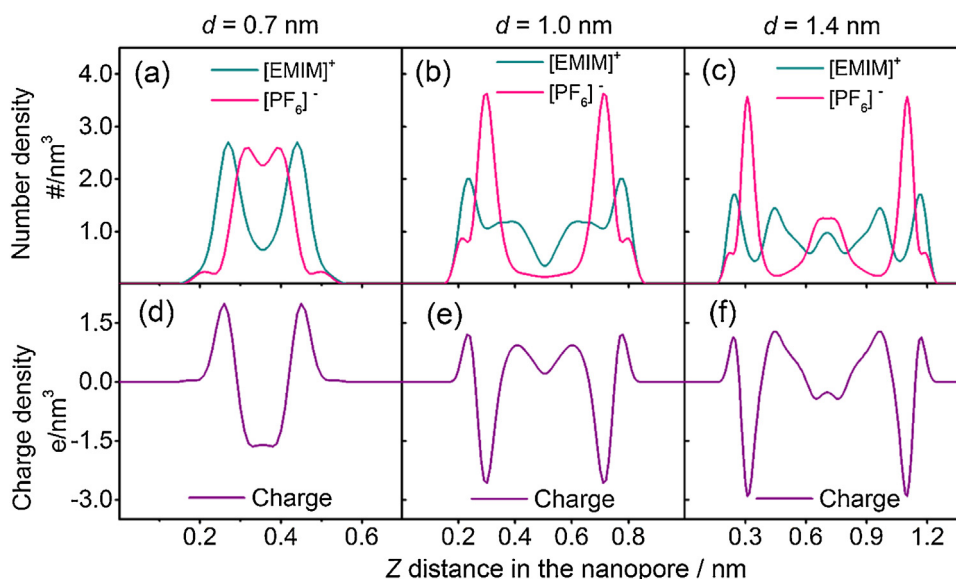


Fig. 5. Number density profile of cation $[\text{EMIM}]^+$ and anion $[\text{PF}_6]^-$ ((a), (b), (c)) and charge density profile ((d), (e), (f)) perpendicular to the $\text{Ti}_3\text{C}_2(\text{OH})_2$ sheet surface in the nanopore with different distance ($d=0.7, 1.0, 1.4 \text{ nm}$).

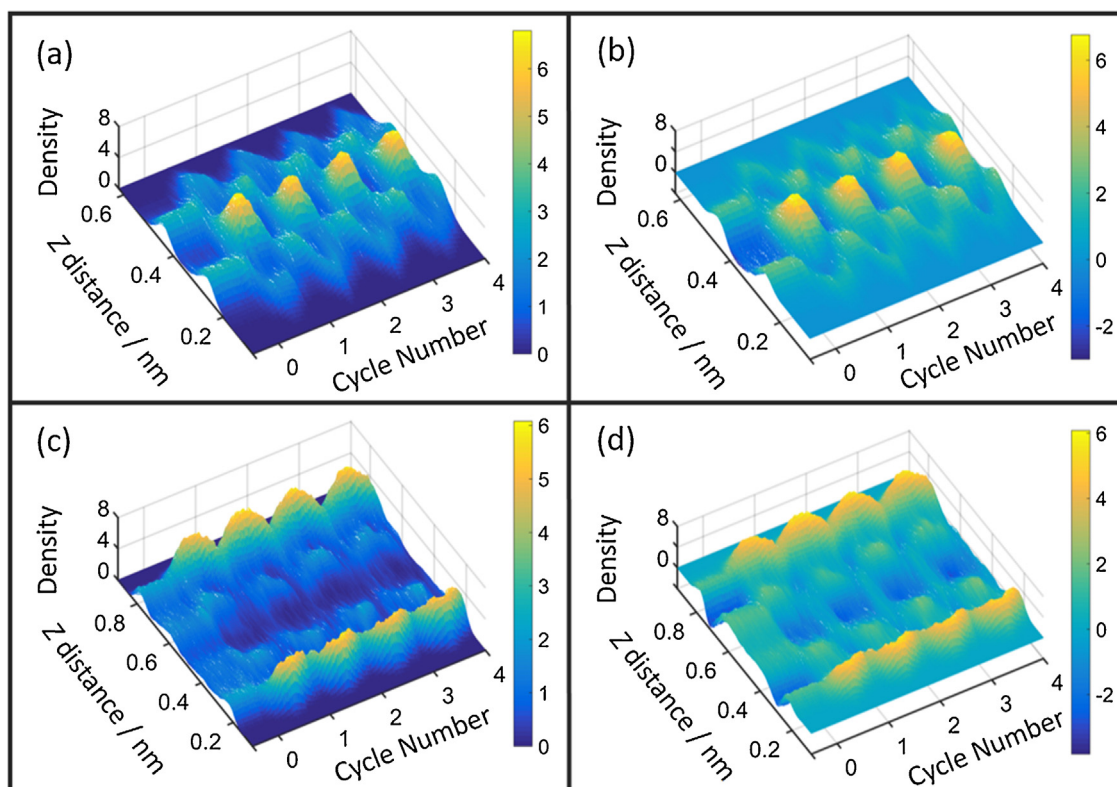


Fig. 6. Structure evolution of cation $[\text{EMIM}]^+$ and charge density profile in cathode pore. (a) and (b) denote the $d = 0.7$ nm pore, (c) and (d) denote the $d = 1.0$ nm pore, $d = 1.4$ nm pore is not shown here, due to its similarity.

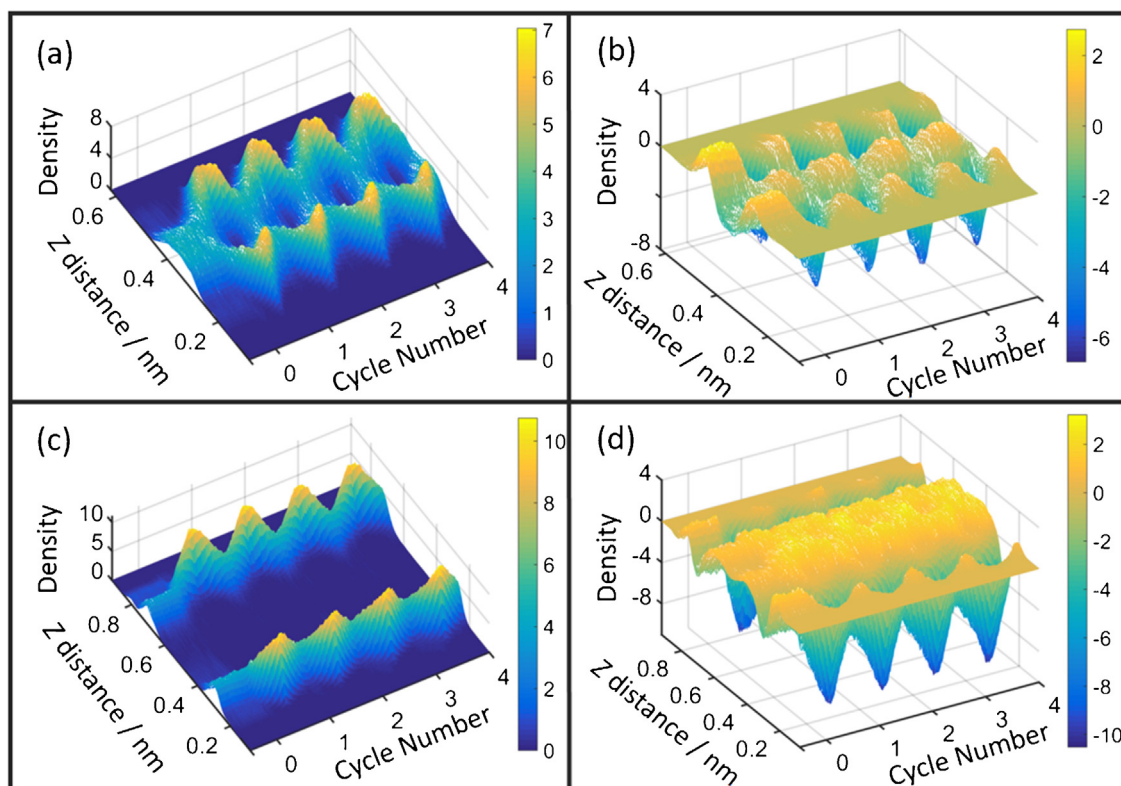


Fig. 7. Structure evolution of anion $[\text{PF}_6]^-$ and charge density profile in anode pore. (a) and (b) denote the $d = 0.7$ nm pore, (c) and (d) denote the $d = 1.0$ nm pore, $d = 1.4$ nm pore is not shown here, due to its similarity.

counter-ions tend to approach and accumulate near the charged electrode surface during charge process. When the discharge begins, this strong accumulation and approaching gradually vanish, and the EDL structure reverts to its neutral state, when the discharge finishes. In the narrow 0.7 nm pores, the profile of the $[\text{EMIM}]^+$ ion exhibiting a peak at the slit center is due to the volumetric limitation of the narrow 0.7 nm pore and the flake-like geometric structure of the $[\text{EMIM}]^+$. This profile differs from other cases forming peaks near the electrode surface. The in-pore charge density profiles exhibit a similar structure to the counter-ion structure at charged state, due to the diminution of co-ion and augmentation of counter-ion in-pore.

3.4. Ion dipole orientation evolution during charge/discharge process

As we all known, there is one more type of responses to the electrostatic interaction exhibited by the polar ion like $[\text{EMIM}]^+$: reorientation of its dipole moment [55]. Insighting into the orientation of the polar $[\text{EMIM}]^+$ ion inside the electrode nanopore offers a clear understanding of above ionic rearrangement phenomenon and a comprehensive understanding of the internal charge storage mechanism. The probability profile of dipole orientation of $[\text{EMIM}]^+$ ion in the electrode pore with different distances are provided in Fig. 8. The θ is defined as the angle between plane direction of the IM-ring and the electrode surface normal (z), as shown in Fig. 8(a). It can be seen that, in the case of $d = 0.7$ nm pore, the dipole orientations is mainly around $20^\circ/160^\circ$, indicates that the IM-ring of $[\text{EMIM}]^+$ in-pore tends to orient perpendicular to the electrode surface at neutral state. As the pore distance increase, the steric confine effect become weak, their orientation becomes more random. The $[\text{EMIM}]^+$ gradually tends to prefer an incline orientation, when the d increase to 1.4 nm.

Under charged state, The $[\text{EMIM}]^+$ ion rotates to help to screen the external electric field, and its orientation of dipole also exhibit a cyclic variation during charge/discharge process, as delineated in Fig. 8(b)(c)(d). For the $d = 0.7$ nm and 1.0 nm pore, the $[\text{EMIM}]^+$ ion exhibit a dipole rotation and turn to prefer parallel and incline orientation. Interestingly, the $[\text{EMIM}]^+$ inside the 1.4 nm pore exhibit an opposite trend, rotates to a perpendicular orientation at

charged state. The different dipole orientation and density distribution at neutral state promote these diverse responses in the pores with different distance. The ion dipole rotation accompanied with the ion rearrangement play an important role in screening the external electric field. And they may exhibit a faster response comparing with the dynamic controlled ion exchange and insertion.

4. Conclusions

To briefly summarize, in this work we theoretically explore the dynamic properties of RTIL electrolyte in the MXene ($\text{Ti}_3\text{C}_2(\text{OH})_2$) electrode featuring intrinsic slit nanopore during charge/discharge process using the classic MD simulation. The simulation results show that the electrolyte ions spontaneously wet the $\text{Ti}_3\text{C}_2(\text{OH})_2$ slit nanopore in the absence of external potential, even at 0.7 nm layer distance pore case. An intriguing electroneutrality breakdown and faster diffusion property is observed inside the nanopore. During charge/discharge process, the charge storage takes place by a fast counter-ion insertion and counter-ion/co-ion exchange. The total ionic number inside the nanopore cyclically oscillates, which may induce a volumetric expansion and shrinkage between the electrode layers. Besides, the ion and net charge spatial distribution perpendicular to the pore surface and the orientation of the cation dipoles inside the pore vary cyclically alone with the charge/discharge process, which helps to screen the external electric potential. Having elucidated the charging mechanism, we investigate the layer distance factor and show that the ions exhibit more significantly acceleration and more ordered dipole orientation inside the narrower pore. This will help $\text{Ti}_3\text{C}_2(\text{OH})_2$ electrode with narrower pore exhibit better power performance. These simulation results give us an atomic insights into the dynamic response of charge storage both spatially and temporally, which is very difficult, if not impossible, to obtain in experimental studies.

It's worth noting that in this work, and also in plenty present atomic simulation studies, the possible chemical processes, such as electronic transfer between electrode and electrolyte and the electronic polarization are neglected. Also the local accumulation

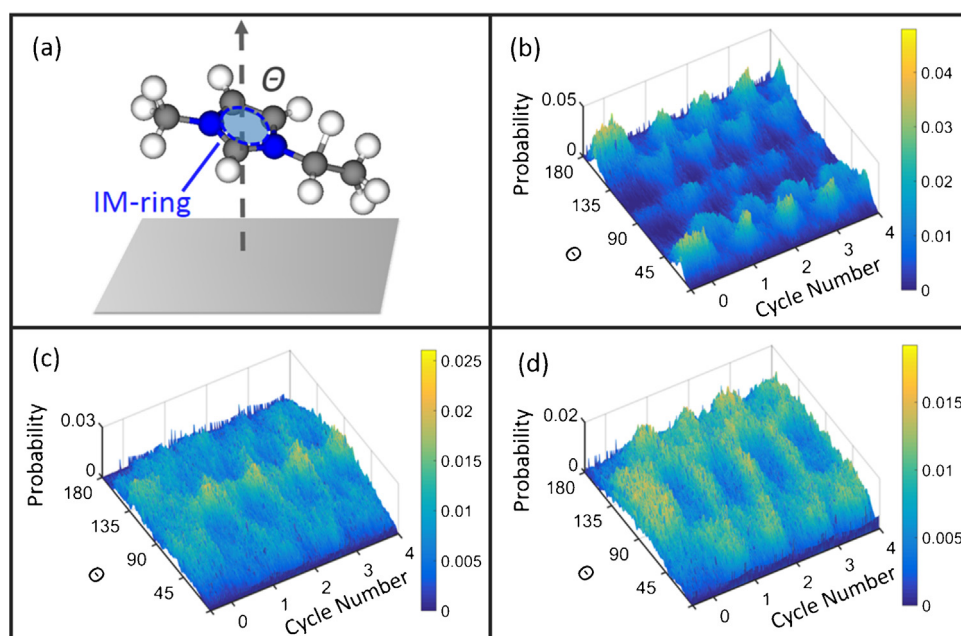


Fig. 8. (a) The schematic of angle θ between IM-ring plane and electrode surface normal. Probability profiles evolution of dipole orientation of $[\text{EMIM}]^+$ inside the $d = 0.7$ nm (b), $d = 1.0$ nm (c) and $d = 1.4$ nm (d) electrode pore.

of charges at the pore entrance is not considered. In practice, these electronic effects play an important role in charge storage. Clarifying these effects requires simulations resolved in multi-scales, including quantum and molecular, which is remaining a big challenge in state-of-art of materials modeling. However, understanding the dynamic charge storage inside the nanopore is fundamental for promoting their electrochemical performance in the field of energy storage systems. We believe that the findings in our work will offer a reasonable guidance for future multiscale simulations incorporating both electronic transfer and electronic polarization effects. Finally, functionalization of MXene electrodes is effected solely via —OH group in our model. Although —OH group is one of predominant functional groups in MXene, other functional groups, especially —O and —F have also been observed experimentally. Since the different electronegativity of the functional groups, different degrees of charge redistributions will occur around the MXene electrode surface. The different surface polarity will significantly impact the arrangement of RTIL ions and the EDL near the electrode surface, in company with the different surface specific adsorption interaction. More details about the role of different functional groups will be investigated in our series work.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.electacta.2016.02.165>.

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